

Introduction to MATLAB

Chapter 2: Plotting with MATLAB

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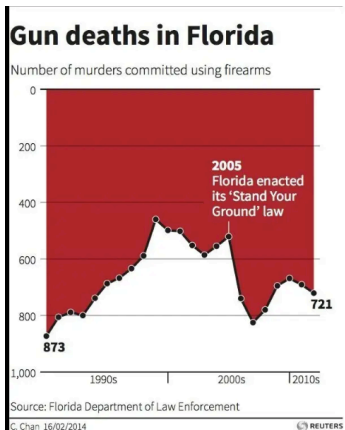
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Introduction

- Plots give a visual understanding of problems; mandatory whatever you end up doing
- Easy to manipulate on MATLAB
- Today's session will be divided in 2 parts:
 1. How to build 2D plots
 2. How to build 3D plots

Example I



Example II

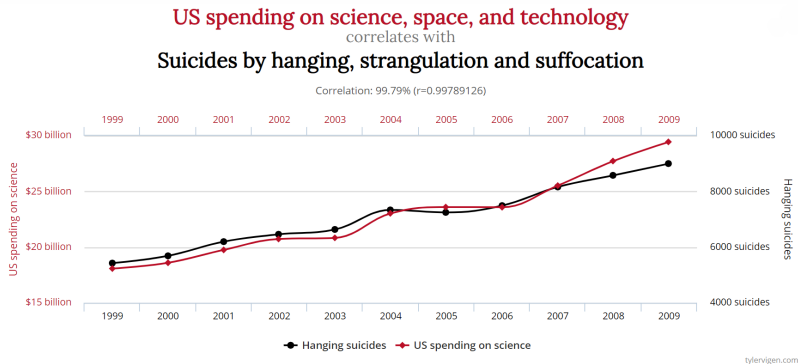


Figure: *

Source: <https://www.maartengrootendorst.com/blog/misleading/>

Simple 2D plot: use `plot(.)`

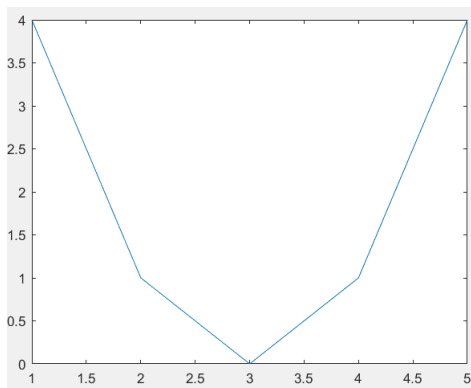
- `plot(.)` is the function used to create 2D plot
- Input: one row or column vector
- Automatically put on the y-axis the value of the vector and on the x-axis the index
⇒ plot all pairs $(1, y_1), \dots, (n, y_n)$
- Connect all the dots with a line

Simple 2D plot : example

1. Create an equally-spaced vector of dimension (1x5)
 2. Create a new vector Y which applies the transformation $f(x) = x^2$ to each element of X
 3. Plot the Y-vector versus the index of each of its value
- (1, y_1), (2, y_2), ... (5, y_5) with $y_1 = 4, y_2 = 1, \dots y_5 = 4$

```
X = -2:2;  
Y = X.^2;  
plot(Y)
```

Simple 2D plot: example



Problem : How to plot $f(x) = x^2$ on the interval $[-2, 2]$?
⇒ Use the function `plot(.,.)`

2D plot with personalized x-axis

- `plot(.,.)` is the function used to create 2D plot with personalized x-axis
- Input: two row or column vectors X and Y
- Plot all pairs $(x_1, y_1), \dots, (x_n, y_n)$
- Connect all the dots with a line

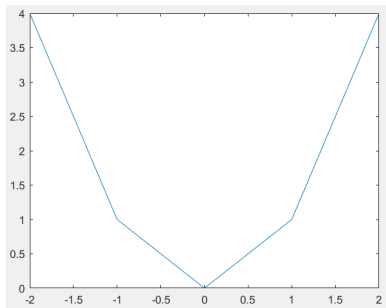
2D plot with personalized x-axis : example

Plot the data in Y versus the corresponding values in X

```
X = -2:2;  
Y = X.^2;  
plot(X,Y)
```

Note: X and Y must be of the same length

2D plot with personalized x-axis : example



Poor graphical representation of $f(x) = x^2$ due to the density of the grid
Try with a denser grid using:

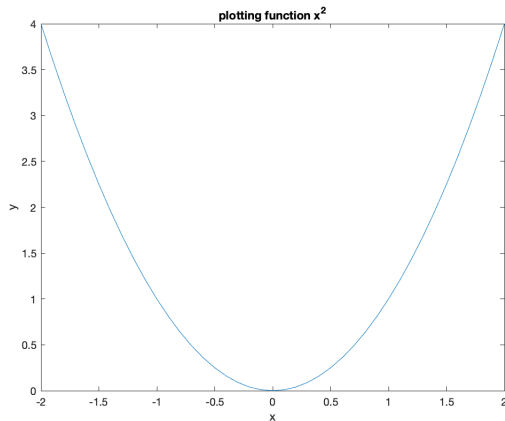
```
X = -2:0.1:2;
```

2D plot with title and labels for the x-axis and the y-axis

Include a title to the plot and name the x and y axis

```
plot(X,Y)
title('plotting function x^2')
xlabel('x')
ylabel('y')
```

2D plot with title and labels for the x-axis and the y-axis



Exercise 1

1. Plot $y = \cos(x)$ on the support $[-10, 10]$.
*Careful: $-10:10$ is only 21 integer points — the support is given, the *step* is your decision.*
2. Provide a title and labels for axis
3. On the same figure, plot $z = 2\cos(x)$ using hold on; command
4. Still on the same figure, plot $w = \cos(2x)$ with a green dashed curve
Tips: use whatever tool you like (an assistant, or help plot) — you must be able to explain the line you end up with.
5. Draw an horizontal line on $f(x)=1$, that one should be dotted.
6. We want to show only values of x above 0. Use xlim to restrict the plotted support.
7. Same question, but for values of y , z and $w > 0$.
Two honest answers: ylim (zoom) and NaN masking (delete). Do both — which is more misleading?
8. Provide a legend

Line in 3D space

- `plot3(x,y,z)` is used to plot lines in 3D space
- One additional argument is required for the additional dimension
- The function plots the values in z as heights above the x - y plane given by x and y

Line in 3D space : example

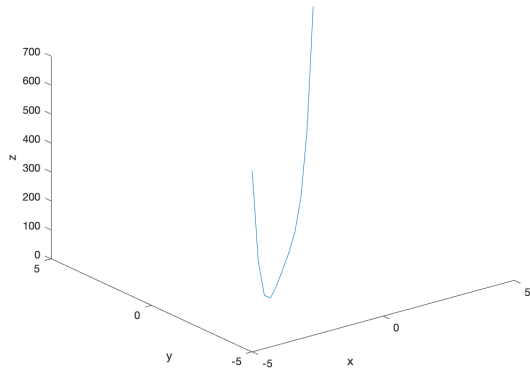
```
x = -5:5;  
y = -5:5;  
z = (x.*y).^2;  
plot3(x,y,z)  
xlabel('x')  
ylabel('y')  
zlabel('z')
```

→ z is a vector composed of 11 elements associated with each pair $(-5,-5), (-4,-4), \dots, (5,5)$

Note 1: x and y are both vectors of the same size, hence z will keep the same size

Note 2: z is not defined for all pairs (x,y) such that $x \neq y$

Line in 3D space : example output



Surface in 3D space

We want to plot $z=f(x,y)$ **for all** possible (x,y) (not just the one such that $x=y$)

- At the difference of a line in 3D, we need a value of z for each pair (x,y)
- X and Y are vectors composed of the elements where the function is evaluated
- Z will be a matrix: for each given x , we need to compute z for all possible y ; and for each given y we need to compute z for all possible x
- To get our matrix Z we need to transform X and Y into matrices

Surface in 3D space : transform vectors into matrices

We want to transform x and y into matrices such that applying the transformation $f(.,.)$ to X and Y yields Z

- The function `meshgrid(x,y)` yields two matrices with the first having the rows filled of copies of the vector x and the second one having the columns filled of copies of the vector y
- Now, applying the transformation $f(.,.)$ to our new X and Y will yield Z for all possible pairs (x,y)

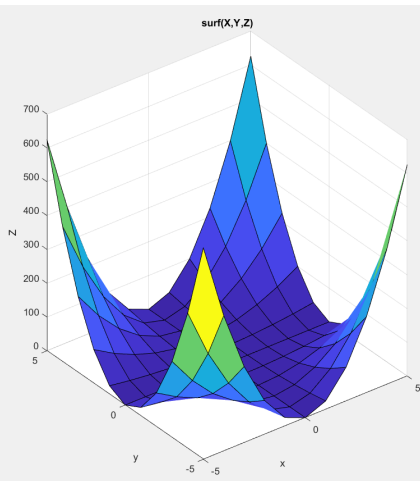
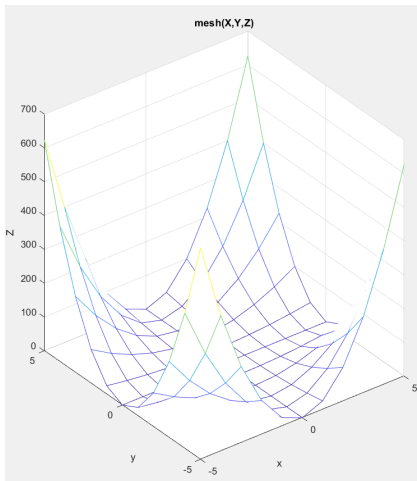
Surface in 3D space : plotting

- The function `mesh(x,y,Z)` plots the values in matrix `Z` as heights above a grid in the `x-y` plane defined by `X` and `Y`
Note: the surface has no color
- The function `surf(x,y,Z)` works the same but now the surface has color

Surface in 3D space : example

```
x = -5:5;
y = -5:5;
[X,Y] = meshgrid(x,y);
Z = (X.*Y).^2;
subplot(1,2,1)
mesh(X,Y,Z)
xlabel('x'); ylabel('y'); zlabel('Z');
title('mesh(X,Y,Z)')
subplot(1,2,2)
surf(X,Y,Z)
xlabel('x'); ylabel('y'); zlabel('Z');
title('surf(X,Y,Z)')
```

Surface in 3D space : example output



The dot that does not error

On a *vector*, forgetting the dot stopped you:

`X = -2:0.1:2; X^2` → *Incorrect dimensions for raising a matrix to a power.*

Same mistake on the *grid* — `(X.*Y).^2` vs `(X.*Y)^2`:

	<code>(X.*Y).^2</code>	<code>(X.*Y)^2</code>
errors?	no	no
value at (1,1)	625	2750
maximum	625	2750

`meshgrid` returns a **square** matrix, so `^2` is legal — it is $M * M$. Two plausible surfaces, no warning.
⇒ *Whether a bug shouts at you is luck, not skill.*

Surface in 3D space : Exercise 2

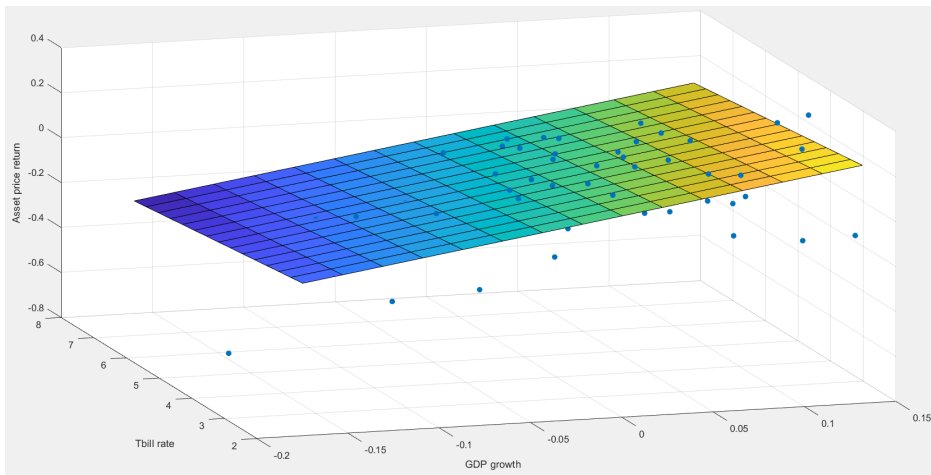
Copy paste the following code: (careful with the underscore — if it fails, type it by hand)

```
load Data_NelsonPlosser
ourData = Data(51:end,:);
Time = dates(51:end);
SP = ourData(:,14);
BY = ourData(:,13);
GNPPC = ourData(:,3);
```

Surface in 3D space : Exercise 2

1. Plot the price return of assets with time in abscissa.
2. Estimate $r_t = b_0 + b_1 x_{1,t} + b_2 x_{2,t} + e_t$ with $x_{1,t} = g_t$ and $x_{2,t} = BY_t$ (how much asset price return is correlated with GDP growth and treasury bond yield).
3. Use the scatter plot function `scatter3(·)` (with 'filled' option) to plot the points.
4. Next line, write "hold on;" to continue the modification of the plot. For x_1 , create a vector of 15 evenly spaced points in the interval $[\min(x_1), \max(x_1)]$ named `x1fit` using the function `linspace(·)`. Same for x_2 with a vector called `x2fit`.
5. Compute $\hat{Z}$ based on estimated parameters.
6. Perform a 3D plot via `mesh()` function with GDP growth on the x-axis and treasury bond yield on the y-axis (use labelling).

Surface in 3D space : Exercise 2 Output



Assignment for next session

1. Load the US macro data (same as exercise 2)
2. Replicate the following figure:

